

Sharp Enclosures for Trace-Normalized Gram Snapshots

A Validated Source Interface for the Directional Packet Program

Prime Dynamics Theory Program

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Abstract

Let S be a nonzero finite matrix and

$$\mathcal{N}(S) = \frac{S^* S}{\|S\|_F^2}.$$

This trace-one positive matrix is the normalized snapshot consumed by the moving-memory construction of RH-130–RH-139. We prove a sharp perturbation theorem for this nonlinear normalization. If $S, H \neq 0$ and

$$\delta = \min \left\{ 1, \frac{\|S - H\|_F}{\max(\|S\|_F, \|H\|_F)} \right\},$$

then

$$\|\mathcal{N}(S) - \mathcal{N}(H)\| \leq \delta, \quad \|\mathcal{N}(S) - \mathcal{N}(H)\|_F \leq \sqrt{2} \delta, \quad \|\mathcal{N}(S) - \mathcal{N}(H)\|_1 \leq 2\delta.$$

All three constants are simultaneously attained by one-row matrices. If H is an exact orthogonal singular-value truncation and $q = \|S - H\|_F / \|S\|_F$, the same bounds improve quadratically to $q^2, \sqrt{2}q^2, 2q^2$, with equality in trace norm. The degeneration as $\delta \uparrow 1$ is unavoidable because normalization is discontinuous at the zero matrix.

We then transport the 160-bit Arb residuals of RH-77 through the universal theorem at all five archived scales and both source channels. Every rank-four postblock snapshot lies in a certified operator ball of radius at most 2.824×10^{-4} ; eight of ten radii are below 1.34×10^{-6} . The direct floating proxies exhibit the expected quadratic cancellation, but we do not treat that cancellation as interval-validated. Thus snapshot normalization is no longer an uncontrolled finite interface. Spectral projector stability, packet construction, threshold updates, and every all-level statement remain open.

1 The interface problem

The source model produces rectangular postblock states, whereas the later directional machinery consumes trace-normalized right Gramians. A small absolute state error alone does not control this map near zero: both numerator and denominator change, and the denominator can vanish. RH-77 supplied Arb balls for the state residual after rank compression, but those

balls had not been converted into snapshot balls. Consequently RH-138 validated a frozen matrix assembly rather than the intended source model.

The correct geometry is projective Hilbert–Schmidt geometry. Regard a matrix as a vector in the tensor product of its row and column spaces. The rank-one density matrix of the normalized vector has $\mathcal{N}(S)$ as a partial trace. Contractivity of trace distance under partial trace then gives constants independent of either matrix dimension [Nielsen and Chuang, 2010]. This dimension-free feature is essential at the five scale-dependent discretizations.

2 Sharp normalized-snapshot enclosure

Write $\|\cdot\|$, $\|\cdot\|_F$, and $\|\cdot\|_1$ for operator, Hilbert–Schmidt, and trace norm. The following theorem is valid over real or complex matrices.

Theorem 1 (Sharp projective snapshot bound). *For nonzero matrices S, H of the same shape, set*

$$d = \|S - H\|_F, \quad \delta = \min \left\{ 1, \frac{d}{\max(\|S\|_F, \|H\|_F)} \right\}.$$

Then

$$\|\mathcal{N}(S) - \mathcal{N}(H)\| \leq \delta, \tag{1}$$

$$\|\mathcal{N}(S) - \mathcal{N}(H)\|_F \leq \sqrt{2} \delta, \tag{2}$$

$$\|\mathcal{N}(S) - \mathcal{N}(H)\|_1 \leq 2\delta. \tag{3}$$

For every $0 \leq \delta < 1$, one pair of one-row matrices attains equality in all three inequalities.

Proof. Let $u = S/\|S\|_F$ and $v = H/\|H\|_F$, interpreted as unit vectors in Hilbert–Schmidt space. If θ is their projective angle, projection orthogonal to either endpoint gives

$$\|S\|_F \sin \theta \leq d, \quad \|H\|_F \sin \theta \leq d.$$

Hence $\sin \theta \leq \delta$. The trace distance between the pure states uu^* and vv^* equals $2 \sin \theta$. Since partial trace is positive and trace preserving on Hermitian matrices, it contracts trace norm, while partial tracing the row factor gives $\mathcal{N}(S)$ and $\mathcal{N}(H)$. This proves (3).

The difference is Hermitian and traceless. Its positive and negative parts therefore each have trace at most δ . Thus its operator norm is at most δ , and the sum of the squared eigenvalues is at most $\delta^2 + \delta^2$, proving (1)–(2).

For sharpness take

$$H = (1, 0), \quad S = (1 - \delta^2, \delta\sqrt{1 - \delta^2}).$$

Then $\|H\|_F = 1$, $\|S - H\|_F = \delta$, and the normalized rows meet at angle $\arcsin \delta$. Their snapshot difference has eigenvalues $\pm \delta$. $\square$

An immediate consequence is the outward Loewner enclosure

$$\mathcal{N}(H) - \delta I \preceq \mathcal{N}(S) \preceq \mathcal{N}(H) + \delta I.$$

No factor depending on row count, column count, or rank occurs.

3 Quadratic cancellation for singular truncation

The general theorem cannot exploit the special geometry of a singular-value truncation. That geometry gives one extra order.

Proposition 2 (Orthogonal split). *Suppose $S = H + R$, $H^*R = 0$, $HR^* = 0$, and $q = \|R\|_F/\|S\|_F$. Then*

$$\|\mathcal{N}(S) - \mathcal{N}(H)\| \leq q^2, \quad \|\mathcal{N}(S) - \mathcal{N}(H)\|_F \leq \sqrt{2}q^2, \quad \|\mathcal{N}(S) - \mathcal{N}(H)\|_1 = 2q^2.$$

In particular this applies to an exact truncated SVD.

Proof. Put $A = H^*H/\|H\|_F^2$ and, when $R \neq 0$, $B = R^*R/\|R\|_F^2$. The cross terms vanish, A and B have orthogonal supports, and

$$\mathcal{N}(S) = (1 - q^2)A + q^2B, \quad \mathcal{N}(H) = A.$$

Thus the difference is $q^2(B - A)$. Its positive and negative parts have trace q^2 , proving the trace identity and the other two bounds. $\square$

The distinction between Theorem 1 and Proposition 2 matters computationally. A numerically returned SVD is not automatically an exact orthogonal split of the Arb state. The linear theorem is therefore the certified statement used below; the quadratic formula is recorded as a diagnostic and as a target for a later interval projector calculation. This avoids silently turning backward error into a spectral theorem [ke Wedin, 1972, Bhatia, 1997].

4 The zero-state obstruction

The strict nonzero condition cannot be removed. As $\delta \uparrow 1$ in the sharp witness, $S \rightarrow 0$ while its normalized row tends to a direction orthogonal to H . At relative radius one, arbitrarily small matrices can produce snapshot operator distance arbitrarily close to one. For any radius larger than one, a ball around H contains small matrices in every orthogonal direction. Hence no reference-sensitive modulus below the full state-space diameter survives across zero.

This is also the correct practical gate. A state enclosure must certify a positive norm before normalization. RH-77 does so at the five anchors; an all-level theorem would have to do so uniformly.

5 Arb anchor audit

The audit reads the exact residual and state-norm balls produced at 160-bit precision by RH-77. It does not reinterpret displayed decimal residuals as exact data: the dependency manifest hashes the original Arb result and its construction script. For each scale, side, and rank, the archived upper $q = \|S - H\|_F/\|S\|_F$ is inserted into Theorem 1. Since the maximum endpoint norm is at least $\|S\|_F$, q is a valid, possibly nonoptimal, operator radius.

All ten rank-four balls are below 10^{-3} and eight are below 10^{-5} ; the two finest-scale channels carry the visible residual. The maximum direct floating proxy distance is 6.44×10^{-8} ,

compression	maximum certified radius	channels below gate	gate
rank one	6.019×10^{-1}	2/10	10^{-1}
rank two	9.568×10^{-2}	6/10	10^{-2}
rank four	2.824×10^{-4}	10/10	10^{-3}

Table 1: Dimension-free operator balls inherited from the RH-77 Arb state residuals. Rank-dependent gates summarize scale, not a theorem premise.

consistent with the quadratic SVD law, but only the larger 2.824×10^{-4} radius is claimed as certified. All thirty direct proxies lie inside their Arb-derived universal balls, and the float SVD trace identity holds to the expected roundoff scale.

6 Consequence and claim boundary

The finite source interface can now be written without an implicit normalization assumption:

$$S \in B_F(H, \varepsilon) \implies \mathcal{N}(S) \in [\mathcal{N}(H) - \delta I, \mathcal{N}(H) + \delta I].$$

This supplies the first component of the source-to-packet enclosure P_E identified by RH-139. It does not supply the whole packet. Eigenprojectors can be unstable when a selected cluster has no gap; polar frames can be unstable at rank loss; threshold decisions can change when a singular value meets the cutoff. The next layer must compare the RH-140 radii with the actual packet spectral gaps and state the corresponding obstruction sharply.

No all-level source norm lower, uniform rank-four compression theorem, controlled-tail tube, normalized-base liminf, Stage A theorem, Hilbert–Polya operator, zeta-zero identification, or Riemann-hypothesis statement is proved here. The result is a rigorous finite bridge from validated source-state residuals to normalized snapshot norm balls.

References

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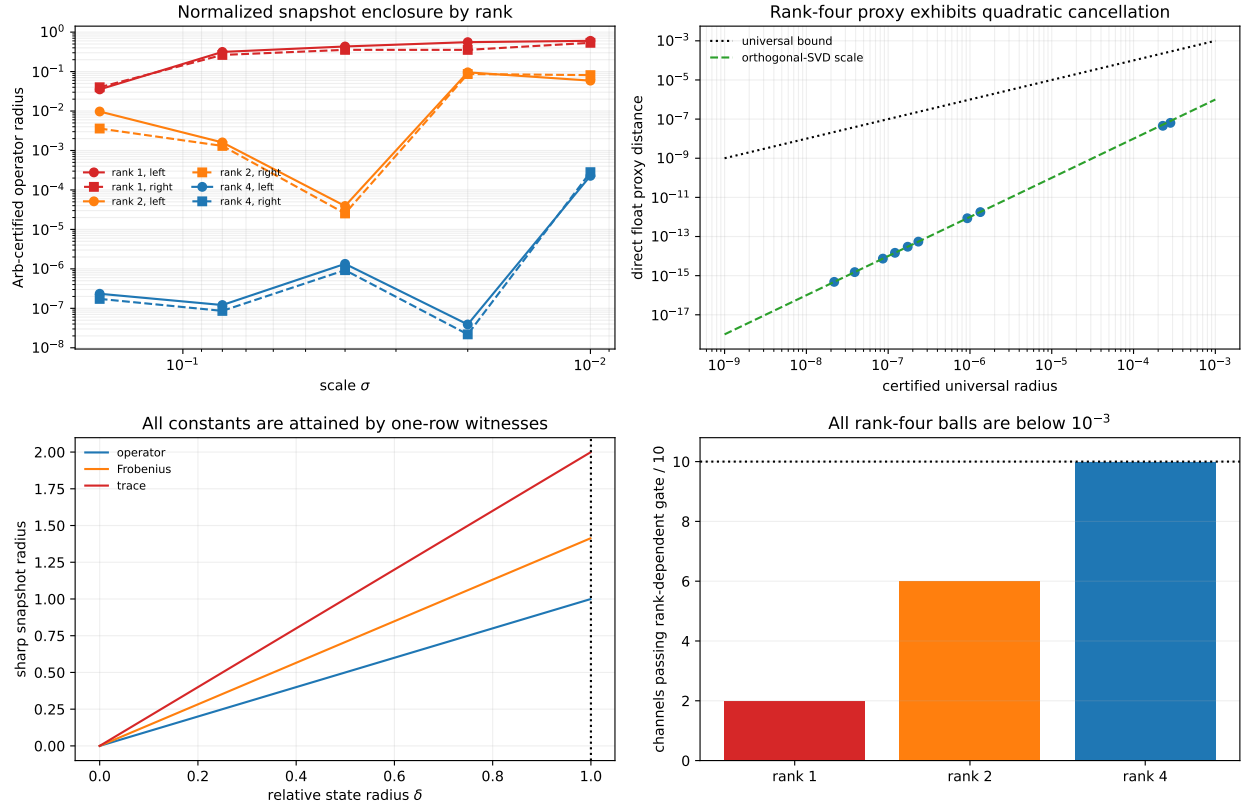


Figure 1: Certified snapshot radii, quadratic cancellation in the rank-four proxy, sharp witness constants, and finite rank-compression gates.